

**A NEW SUBSPECIES OF *VIRACHOLA SMILIS* (HEWITSON)
(LEPIDOPTERA: LYCAENIDAE) FROM
NORTHERN AUSTRALIA**

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Abstract

Virachola smilis dalyensis ssp. n. is described from the Daly River Crossing, south of Darwin, and from Stuart Point, east of Darwin, in the Northern Territory. This is the first record of *V. smilis* (Hewitson) from Australia.

INTRODUCTION

During a visit to the Northern Territory and East Kimberleys from May 26 to July 30, 1969, one of us (J. LeS.) collected an adult male *Virachola* visiting flowers of a cashew nut growing at Daly River Crossing, about 90 miles south of Darwin. The next day another male and a female were taken in a nearby cashew plantation and several other specimens were noticed either in flight or at rest on the flowers.

These specimens have now been identified as *Virachola smilis* (Hewitson), 1863, after an examination of specimens of this genus from India in the U.S. Carnegie Museum and comparison with specimens in the British Museum (Natural History). The previous reported range of *V. smilis* is from Burma to Malaya, the Andaman Islands and Borneo (Corbet and Pendlebury 1956). The discovery of a representative of this species in northern Australia adds considerably to its known distribution.

When this manuscript was in preparation Mr. J. W. C. d'Apice of Sydney collected three adults and reared another from a pupa at Stuart Point, 80 miles east of Darwin.

***Virachola smilis dalyensis* ssp. n.**

Types.—*Holotype* male and *allotype* female labelled "Daly R., N.T., 30 June 1969, J. C. Le Souef"; one *paratype* male with similar label data but collected "29 June 1969", three *paratype* males labelled "Shady Camp, Pt. Stuart, N.T., 21 July 1970, J. d'Apice", one female *paratype* with similar label data but emerged "ex pupa, 28.vii.1970". *Holotype* male deposited in the National Museum of Victoria (T-4203); *allotype* and one *paratype* in the J. C. LeSouef collection, three *paratype* males and one *paratype* female in the J. W. C. d'Apice collection.

Holotype male (Plate 1A, 1B)

Head dark grey, white band behind eye, frons white, labial palpus with second segment white, dull black above in distal half, apical segment dull black, antenna dull black, shaft laterally narrowly banded white, club tipped dull orange-red. Thorax and abdomen dull black, clothed with shining bluish grey pubescence and hairs.

Forewing length 20 mm. Above dark greyish black, discal cell and small area between end of cell and costa velvet black, a broad central area extending from base and lower half of cell to dorsum and nearly to tornus rich shining blue, termen black, cilia tipped ochreous.

Hindwing above dark greyish black, a dull grey rounded sex-mark at base of Rs, inner margin grey, a broad central area rich shining blue with purplish tints, its anterior margin from base across lower half of cell and between M₁ and M₂, not reaching termen but extending to near tornus and bordered by 1A + 2A, termen and veins beyond cell nearly black, terminal line black, cilia whitish distally, tornal lobe pale ochreous, with a large black spot, outwardly edged whitish and dusted with pale metallic green, especially at base, two obscure black subterminal spots between CuA₁ and 1A + 2A, a thin black tail tipped with white at end of CuA₂.

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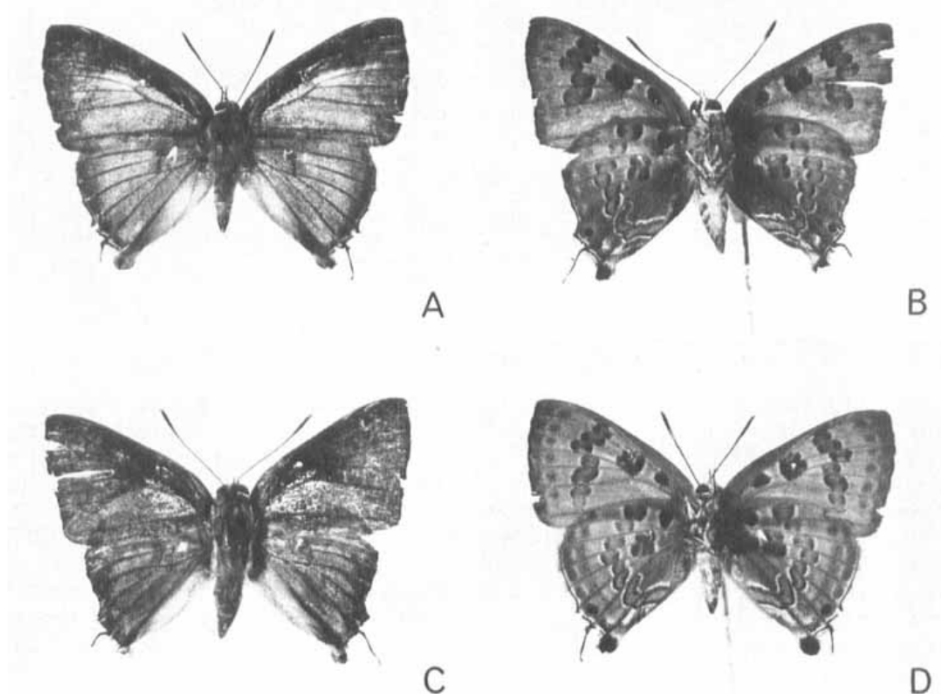
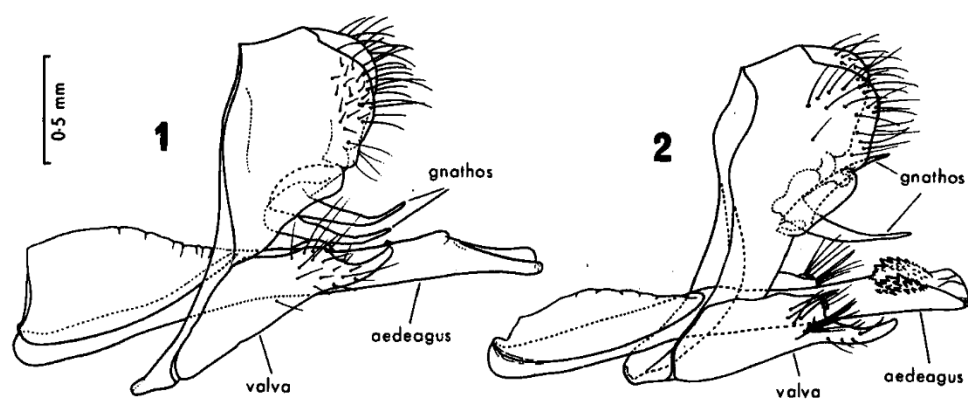


PLATE I

Adults of *Virachola smilis dalyensis* ssp. n.: (A,B) above and beneath holotype male;
(C,D) above and beneath allotype female.



FIGS. 1,2.—Male genitalia of *Virachola*: (1) *V. smilis dalyensis* ssp. n., paratype, Daly River, N.T.;
(2) *V. democles* (Miskin), Cairns, Q.

Forewing beneath dull ochreous grey shading to grey-brown on costa and termen, with a slight purplish tint, a reddish brown spot in cell, a larger grey-brown spot at end of cell bordered dark brown edged whitish, a postmedian band of similar but less distinct spots from R_2 to CuA_2 and very obscurely continued to $1A + 2A$, band clearly broken at M_3 , a very obscure subterminal band of pale spots.

Hindwing beneath dull ochreous grey in costal half shading to grey-brown towards tornus, with a slight purplish tint, a prominent reddish brown subbasal spot between $Sc + R_1$ and cell, a double dark brown bar at end of cell edged whitish, a strongly curved somewhat irregular postmedian band of ochreous grey spots from $Sc + R_1$ to $3A$, obscurely bordered dark brown edged whitish anterior to CuA_1 , then more distinctly bordered brown-black edged slightly metallic bluish white, with a similar spot on inner margin in basal half, a series of obscure whitish subterminal spots, a dark brown terminal line becoming black towards tornus, an ochreous grey subterminal spot enclosing a black spot between CuA_1 and CuA_2 , a black subterminal spot almost obscured by pale metallic green between CuA_2 and $1A + 2A$, a large black spot in tornal lobe basally edged white.

Allotype female (Plate 1C, 1D)

Head as in male but antenna shorter. Forewing length 19 mm, costa more strongly arched and termen more convex than in male, above similar to male but with grey-black margins a little paler, central area pale blue, a small whitish patch beyond end of cell. Hindwing with termen more uniformly convex than in male, above as in male but costa and termen broadly brown-black, central area a duller blue, with a very slight whitish suffusion beyond cell between M_1 and M_2 , a blackish bar at end of cell, veins brown-black, terminal line blackish inwardly edged whitish, cilia whitish, tornal lobe with black spot dusted at base with only a few pale metallic green scales, subterminal spots between CuA_1 and $1A + 2A$ more distinct than in male. Beneath similar to male but ground colour paler, markings paler and less distinct, fore- and hindwing with subterminal and terminal rows of grey-brown spots, tornal lobe with more prominent black spot.

Male genitalia (Fig. 1)

Tegumen hood-like; socii a pair of prominent hairy lobes; gnathos a pair of tapering acute hooks; valva small, broad at base, tapering to a slightly upcurved apex, costa with a rounded median prominence; aedeagus with oblique orifice and rounded apex.

Comments

The Australian specimens have the generic characters of *Virachola*, including the rounded sex-mark on the hindwing above, and a corresponding tuft of long hairs on the forewing beneath. These secondary sexual characters are also found in the closely related genus *Rapala* Moore, but are absent in *Deudorix* Hewitson. Unlike both *Rapala* and *Deudorix*, Sc and R_1 are nearly parallel in the forewing and R_{4+5} and M_1 are very short-stalked from the upper angle of the cell (Corbet and Pendlebury 1956).

V. smilis was regarded by Seitz (1924-1927) as a subspecies of *V. perse* (Hewitson), but the two may be distinguished by the postmedian band beneath the forewing, which is continuous in *V. perse* but clearly broken at M_3 in *V. smilis* (Evans 1932). Further, a spot in the discal cell beneath the forewing is rarely present in *V. perse* but usually present in *V. smilis*.

Typical *smilis* from Burma, and subspecies *maseas* Fruhstorfer from the Andamans, are without the orange, ochreous or white patch at the end of the discal cell above the forewing, found in the various subspecies of *V. perse*. However, in this respect *dalyensis* bears some resemblance to *V. perse*, with a small whitish patch in the forewing of the female. Mr. C. E. Tite states that there are two specimens in the British Museum that resemble *dalyensis*, a male from Batchian and a female from Obi, both localities in the Moluccas.

V. smilis is rather similar to *V. democles* (Miskin) from north-eastern Australia, but the ground colour beneath is much darker and the markings more distinct. In *V. democles* the forewing is without a distinct cell-spot beneath, and the postmedian band is continuous though slightly elbowed at M_2 . The hindwing is also without a distinct subbasal spot between $Sc + R_1$ and the discal cell, and there is only a faint double bar at the end of the cell. The male genitalia of *V. smilis dalyensis* (Fig. 1) lack the distal dorsal dentation of the aedeagus found in *V. democles* (Fig. 2) and the costal prominence of the valva is much more gently rounded.

The pupa of *dalyensis*, from which the paratype female emerged, was found in a hollowed-out fruit of *Strychnos lucida* (Loganiaceae). The larvae apparently feed within the small orange fruits of *Strychnos* and finally pupate within, the rind becoming hard and brittle. The adult escapes from the fruit through a round hole previously made in the rind by the larva. Mr. J. W. C. d'Apice also collected three of the male paratypes flying around a tree next to the food plant. At the type locality adults of *dalyensis* were difficult to distinguish in flight from *Hypolycaena phorbas ingura* Tindale, several of which were flying around the same blossoms.

The other subspecies of *V. smilis* are relatively rare insects and there appears to be no record of their life history. According to Woodhouse (1949) and other authors, the larvae of the related *V. perse* feed within the fruits of *Randia* (Rubiaceae) and guava. The larvae of *V. democles* feed in the fruits of *Strychnos bancroftiana* in northern Queensland.

ACKNOWLEDGEMENTS

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REFERENCES

- CORBET, A. S. and PENDLEBURY, H. M. (1956).—“The Butterflies of the Malay Peninsula”. 2nd Edition. (Oliver & Boyd: Edinburgh).
EVANS, W. H. (1932).—“The Identification of Indian Butterflies”. 2nd Edition. (The Diocesan Press: Madras).
SEITZ, A. (1924-1927).—Lycaenidae. *Gross-Schmetterl. Erde* 9: 901-1010.
WOODHOUSE, L. G. O. (1949).—“The Butterfly Fauna of Ceylon”. 2nd Edition. (The Colombo Apothecaries' Co. Ltd.: Colombo).